

1. Define a data structure and mention two real-world

applications
Data Structures Definition

Data Structure is a way of organizing and retrieving data in such a way that we can perform operations on these data in an effective way.

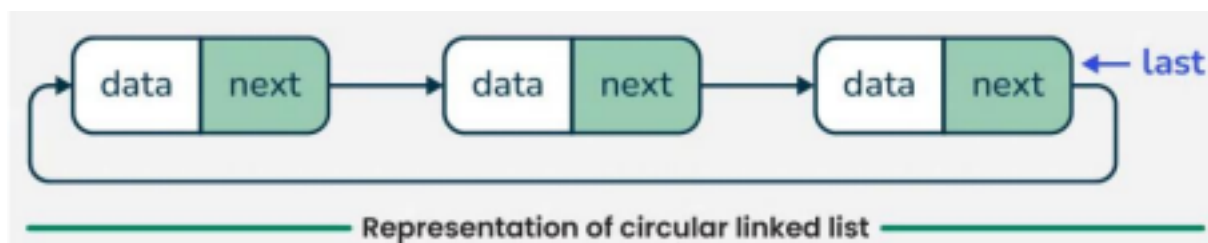
Real-World Applications:

E-commerce: Sorting, search, recommendation systems (Amazon, eBay).

Social Networks: Graph algorithms, BFS/DFS for friend suggestions (Facebook, LinkedIn).

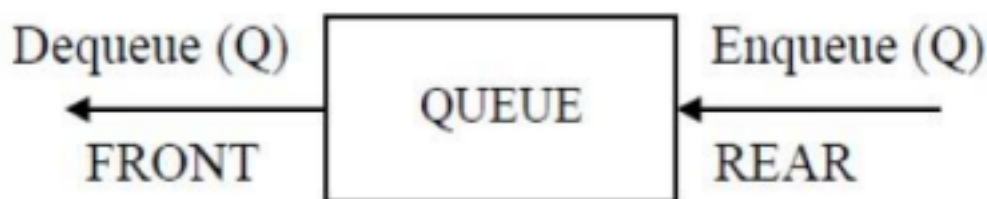
2. What is a circular linked list?

Circular linked list is similar to normal linked list except that the last node contains a pointer to the first node of the list.



3. Define a queue and mention two real-world applications

A Queue is a linear data structure which follows First In First Out (FIFO) principle, in which insertion is performed at rear end and deletion is performed at front end.



Real World Applications

1. Operating Systems & Process Scheduling
2. Task Scheduling in Applications

4. Differentiate Recursion and Iteration.

Aspect Recursion Iteration

Definition in DS

A technique where a function solves a

problem by calling itself, (for, while) repeatedly often execute

A method where loops

Aspect Recursion Iteration

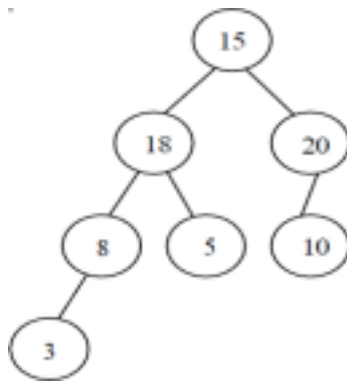
used to traverse or process
recursive data structures
(trees, graphs).

instructions until a condition is
satisfied.

5. Define a binary tree. What is the maximum number of children a binary tree node can have?

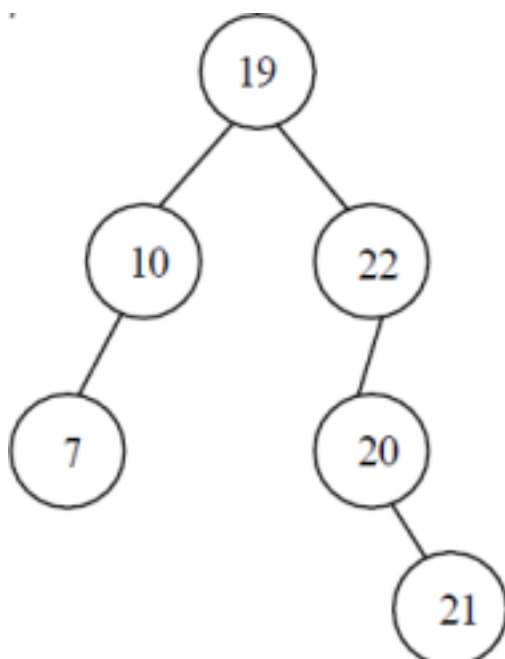
- Binary Tree is a tree in which no node can have more than two children. •

Maximum number of nodes in the tree with level i is $2^i - 1$. (Assume root node is at level 1)



6. What is a binary search tree (BST)?

In a Binary search tree, the value of left node must be smaller than the parent node, and the value of right node must be greater than the parent node. This rule is applied recursively to the left and right subtrees of the root.



7. What is the difference between a min-heap and a max-heap? **Aspect Min Heap Max Heap**

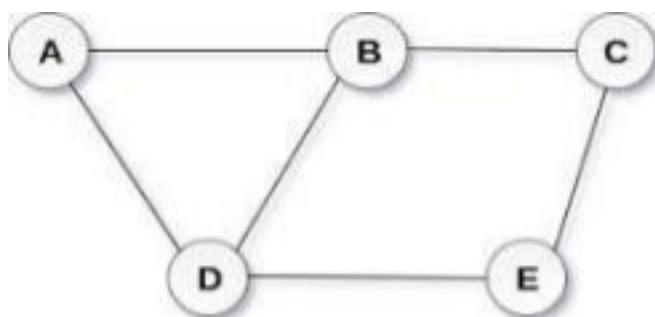
Definition	parent node is always less than or equal to its children.	parent node is always greater than or equal to its children.
Root Node	The minimum element is always at the root.	The maximum element is always at the root.
Node	A binary heap where the	
	upwards if smaller than parent. Insert at bottom → bubble upwards if greater than parent.	

Insertion Insert at bottom → bubble

8. What is a trie in data structures? Name two applications of a trie.

A Trie (pronounced "try", also called a prefix tree) is a tree-like data structure used to store and efficiently retrieve strings, especially when dealing with large sets of words. Each node represents a character of a string. The path from the root to a node represents a prefix. A special marker (often isEndOfWord) is used to indicate the end of a word.

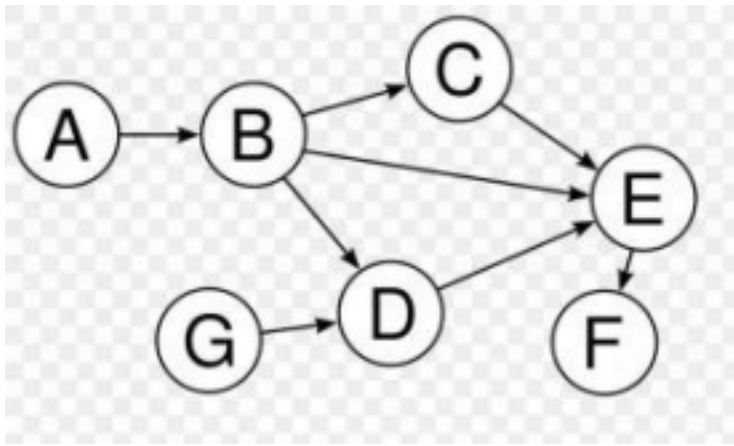
9. What is the difference between a directed and an undirected graph? In an undirected graph, edges do not have any direction associated with them.



Undirected Graph

DIRECTED GRAPHS

In a directed graph, edges have direction associated with them. If there is an edge from A to B, then there is a path from A to B but not from B to A.



10. Give an example of a real-world problem represented by a graph. Google Maps / GPS Navigation

Airline Routes: Airports are nodes, flights are edges with cost/duration as weights.

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11. Outline the linear and nonlinear data structure with an example LINEAR DATA STRUCTURES

If the elements of a data structure are stored in a linear or sequential order, then it is a linear data structure. Examples include arrays, linked lists, stacks, and queues.

NON-LINEAR DATA STRUCTURES

If the elements of a data structure are not stored in a sequential order, then it is a non-linear data structure. The relationship of adjacency is not maintained between elements of a non-linear data structure. Examples include trees and graphs.

12. Convert the infix expression $(a+b)*(c-d)$ to postfix expression $ab+cd-*$

13. List the limitation of linear Queue

- When elements are dequeued (removed) from the front, those spaces cannot be reused.
- Even if there is free space at the front, new elements cannot be inserted

once the rear reaches the end of the array.

14. Write the Overflow and Underflow Conditions in stack

Overflow happens when we try to **push (insert)** an element into a stack that is already **full**.

if (top == size - 1) → Stack Overflow

Underflow happens when we try to **pop (delete)** an element from a stack that is already **empty**.

if (top == -1) → Stack Underflow

15. Construct Binary tree for the following elements

{5,3,7,12,15,23,21,46,20} 5

```

/\
3 7
 \
 12
 \
 15
 \
 23
 /\
21 46
/
20
```

16. What is a binary heap? Outline the properties of a binary heap? BINARY HEAP

The efficient way of implementing priority queue is Binary Heap. Binary heap is merely referred as Heaps

Heap have two properties namely

- Structure property
- Heap order property.

17. List the types of rotation performed on AVL tree with diagram

18. What is a Minimum spanning tree in data structures? Name two applications of a minimum spanning tree.

19. Illustrate the Dijkstra's Algorithm with example

The general method to solve the single source shortest path problem is known as Dijkstra's algorithm. It applied to weighted graph.

Procedure

- It uses greedy technique.
- It proceeds in stages.
- It selects a vertex v , which has the smallest d_v among all the unknown vertices and declares the shortest path from s to v is known.
- The remainder consists of updating the value of d_w .
- We should set $d_w = d_v + C_{v,w}$, if the new value for d_w would an improvement.

20. Define Linked List

Linked list consists of a series of structures, which are not necessarily adjacent in memory. Each structure contains the element and a pointer to a structure containing its successor. We call this the Next Pointer. The last cell's Next pointer points to NULL.

21. State the different types of linked lists

The different types of linked list include singly linked list, doubly linked list and circular linked list.

22. List out the disadvantages of using a linked list

- Searching a particular element in a list is difficult and time consuming
- A linked list will use more storage space than an array to store the same number of elements

23. State the difference between arrays and linked lists

Arrays	Linked Lists
Size of an array is fixed	Size of a list is variable
It is necessary to specify the number of elements during declaration.	It is not necessary to specify the number of elements during declaration
Insertions and deletions are somewhat difficult	Insertions and deletions are carried out easily
It occupies less memory than a linked list for the same number of elements	It occupies more memory

24. What are the types of queues?

- Linear Queues – The queue has two ends, the front end and the rear end. The rear end is

where we insert elements and front end is where we delete elements. We can traverse in a

linear queue in only one direction ie) from front to rear.

- Circular Queues – Another form of linear queue in which the last position is connected

to the first position of the list. The circular queue is similar to linear queue has two ends,

the front end and the rear end. The rear end is where we insert elements and front end is

where we delete elements. We can traverse in a circular queue in only one direction ie)

from front to rear.

- Double-Ended-Queue – Another form of queue in which insertions and deletions are made at both the front and rear ends of the queue. 25.

List the applications of stacks

- Towers of Hanoi
- Reversing a string
- Balanced parenthesis

- Recursion using stack
- Evaluation of arithmetic expressions

26. Define a stack

Stack is an ordered collection of elements in which insertions and deletions are restricted to one end. The end from which elements are added and/or removed is referred to as top of the stack. Stacks are also referred as piles, push-down lists and last-in-first-out (LIFO) lists.

27. Define a tree

A tree is a collection of nodes. The collection can be empty; otherwise, a tree consists of a distinguished node r , called the root, and zero or more nonempty (sub) trees $T_1, T_2, \dots, T_k$, each of whose roots are connected by a directed edge from r .

28. Define depth and height of a node

For any node n_i , the depth of n_i is the length of the unique path from the root to n_i . The height of n_i is the length of the longest path from n_i to a leaf.

29. Define a full binary tree

A full binary tree is a tree in which all the leaves are on the same level and every non-leaf node has exactly two children.

30. Define a complete binary tree

A complete binary tree is a tree in which every non-leaf node has exactly two children not necessarily to be on the same level.

31. Define Graph.

A graph G consists of a nonempty set V which is a set of nodes of the graph, a set E which is the set of edges of the graph, and a mapping from the set of edges E to a set of pairs of elements of V . It can also be represented as $G=(V, E)$.

32. What is meant by strongly connected in a graph?

An undirected graph is connected, if there is a path from every vertex to every other vertex. A directed graph with this property is called strongly connected.